

UDIT RAJ KASHYAP

+91 9810547115 | uditrajkashyap442@gmail.com | linkedin.com/in/udit-r-kashyap-828a5a320
github.com/uditrajkashyap442-beep

PROFESSIONAL SUMMARY

Third-year Electronics and Communication Engineering student at VIT Vellore, pursuing a B.Tech in Electronics and Communication Engineering. I have hands-on experience across full-stack development, computer vision, IoT, and embedded systems, and enjoy working on projects that bring software and hardware together. On the software side, I have built and shipped mobile apps using React Native and Expo, set up backend services with Java Spring Boot and FastAPI, and worked with PostgreSQL for data storage. On the hardware side, I have worked with microcontrollers like the ESP32 and 8051, written firmware in C++, and built sensor systems using I2C peripherals. I am comfortable picking up new tools and technologies as needed, and I enjoy the process of taking a project from an idea to something that actually works. Currently looking for opportunities where I can contribute to real-world engineering problems and keep growing as a developer.

EDUCATION

Vellore Institute of Technology (VIT) — Vellore, Tamil Nadu, India | B.Tech in Electronics and Communication Engineering (Core)
| 2024 – Present

Relevant Coursework: Digital Systems and Design · Electronic Materials and Devices · Sensors · Python · MATLAB · Analog Circuits · Signals and Systems · Probability and Statistics

TECHNICAL SKILLS

Programming & Analysis: Python 3, NumPy, Statistical Modeling, R Programming

Full Stack & Backend: Java 21, Spring Boot, Node.js, Flask, RESTful APIs, Microservices, PostgreSQL, Spring Data JPA

Mobile Development: React Native, Expo, TypeScript, Android APK Build, Expo Application Services (EAS)

Embedded & Hardware: C++, Verilog HDL, ESP32, ESP32-CAM, 8051 Microcontroller, I2C, SPI, FFT Analysis, ADXL345, HX711

Frontend & Visualization: HTML5, CSS3, JavaScript, Plotly, Matplotlib

DevOps & Tools: Git, CI/CD Pipelines, Agile, System Design, Cloud Basics

TECHNICAL PROJECTS

Kamikatsu Recycler – Waste Disposal Guide App (2026)

React Native · Expo · TypeScript · Java 21 · Spring Boot · PostgreSQL · Google Gemini AI · REST APIs · CI/CD · EAS

- Built a mobile app (React Native + Expo) that helps users figure out how to properly dispose of products, and packaged it as an Android APK using Expo Application Services.
- Set up a Java / Spring Boot backend with a PostgreSQL database storing 500+ products, recycling categories, and user scan history.
- Added a barcode lookup system that pulls product info from Open Food Facts / Open Beauty Facts APIs and falls back to Google Gemini AI, achieving around 90% recognition across 10+ markets.
- Worked through Android build issues and dependency conflicts to bring the build failure rate down to near zero.

Bridge Structural Health Monitoring Node (2025)

Team contribution — member of a 3-person team

ESP32 · C++ · FFT Analysis · ADXL345 · HX711 · I2C

- Worked on a sensor node that uses FFT to track a bridge's vibration patterns and flag changes that might point to structural wear or damage.
- Connected an ADXL345 accelerometer and HX711 load cell over I2C to capture vibration and load data at a 200 Hz sampling rate.
- Used frequency-domain analysis to spot anomalies early, giving faster warnings compared to manual inspection cycles.

Solar MPPT Charge Controller (2025)

Team contribution — member of a 3-person team

Arduino · C++ · DC-DC Boost Converter · PWM · ADC · I2C LCD · Embedded Systems

- Built a solar charge controller that uses the Perturb and Observe (P&O;) algorithm to continuously track the Maximum Power Point (MPP) of a solar panel, improving energy harvest compared to a direct connection.
- Used high-frequency PWM via the TimerOne library to control a custom DC-DC boost converter, enabling dynamic adjustment of the operating voltage in real time.
- Implemented ADC averaging to filter out electrical noise from voltage and current sensors, giving cleaner and more stable power readings.
- Added multi-stage battery charging (Bulk and Float states) and a 16x2 I2C LCD to display live power stats and charging status.

AREAS OF INTEREST

Internet of Things (IoT) · Computer Vision · Embedded Systems · Full Stack Development · Digital System Design · Signal Processing · Machine Learning